

Introduction to Particle Physics

Quantum Electrodynamics

Homework: Read Section 7.1 - 7.3 of Griffiths.

The Photon

In classical electrodynamics, the electric and magnetic fields (**E** and **B**) produced by a charge density ρ and a current density **J** are determined by Maxwell's equations:*

$$\left\{ \begin{array}{ll} \text{(i)} & \nabla \cdot \mathbf{E} = 4\pi\rho \\ \text{(ii)} & \nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} = 0 \\ \text{(iii)} & \nabla \cdot \mathbf{B} = 0 \\ \text{(iv)} & \nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} = \frac{4\pi}{c} \mathbf{J} \end{array} \right\}$$

In relativistic notation, **E** and **B** together form an antisymmetric second-rank tensor, the 'field strength tensor', $F^{\mu\nu}$:

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix}$$

(that is, $F^{01} = -E_x$, $F^{12} = -B_z$, etc.), while ρ and **J** constitute a four-vector:

$$J^\mu = (c\rho, \mathbf{J})$$

The inhomogeneous Maxwell equations, (i) and (iv) can be written more neatly in tensor notation

$$\partial_\mu F^{\mu\nu} = \frac{4\pi}{c} J^\nu$$

From the antisymmetry of $F^{\mu\nu}$ ($F^{\nu\mu} = -F^{\mu\nu}$) it follows that J^μ is divergenceless:

$$\partial_\mu J^\mu = 0$$

Or, in three-vector notation, $\nabla \cdot \mathbf{J} = -\partial\rho/\partial t$; this is the 'continuity equation', expressing local conservation of charge

As for the homogeneous Maxwell equations, (iii) is equivalent to the statement that **B** can be written as the curl of a vector potential, **A**:

$$\mathbf{B} = \nabla \times \mathbf{A}$$

With this, (ii) becomes

$$\nabla \times \left(\mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{A}}{\partial t} \right) = 0$$

which is equivalent to the statement that $\mathbf{E} + (1/c)(\partial\mathbf{A}/\partial t)$ can be written as the gradient of a scalar potential, V :

$$\mathbf{E} = -\nabla V - \frac{1}{c} \frac{\partial \mathbf{A}}{\partial t}$$

In relativistic notation,

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

where

$$A^\mu = (V, \mathbf{A})$$

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In terms of this four-vector potential, the inhomogeneous Maxwell Equations read:

$$\partial_\mu \partial^\mu A^\nu - \partial^\nu (\partial_\mu A^\mu) = \frac{4\pi}{c} J^\nu$$

The *defect* of the potential formulation is that V and $\mathbf{A}$ are not uniquely determined. Indeed, $\mathbf{A}_{\text{new}}$ potentials

$$A'_\mu = A_\mu + \partial_\mu \lambda$$

(where λ is any function of position and time) would do just as well, since $\partial^\mu A^{\nu'} - \partial^\nu A^{\mu'} = \partial^\mu A^\nu - \partial^\nu A^\mu$. Such a change of potentials, which has no effect on the fields, is called a *gauge transformation*. We can exploit this gauge freedom to impose an extra constraint on the potential

$$\partial_\mu A^\mu = 0$$

In QED, A^μ becomes the wave function of the photon. The *free* photon satisfies

$$\square A^\mu = 0$$

which we recognize in this context as the Klein–Gordon Equation for a massless particle. As in the case of the Dirac equation, we look for plane-wave solutions with four-momentum $p = (E/c, \mathbf{p})$:

$$A^\mu(x) = a e^{-(i/\hbar)p \cdot x} \epsilon^\mu(p)$$

Here ϵ^μ is the *polarization vector* – it characterizes the spin of the photon – and a is a normalization factor.

Lorentz condition requires that

$$p^\mu \epsilon_\mu = 0$$

In the Coulomb gauge, moreover,

$$\epsilon^0 = 0, \quad \text{so} \quad \boldsymbol{\epsilon} \cdot \mathbf{p} = 0$$

which is to say that the polarization three-vector ($\boldsymbol{\epsilon}$) is perpendicular to the direction of propagation; we say that a free photon is *transversely polarized*.* Now, there are *two* linearly independent three-vectors perpendicular to $\mathbf{p}$; for example, if $\mathbf{p}$ points in the z direction, we might choose

$$\boldsymbol{\epsilon}^{(1)} = (1, 0, 0), \quad \boldsymbol{\epsilon}^{(2)} = (0, 1, 0)$$

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Electrons

$$\psi(x) = ae^{-(i/\hbar)p \cdot x} u^{(s)}(p)$$

where $s = 1, 2$ for the two spin states. The spinors $u^{(s)}$ and $v^{(s)}$ satisfy the momentum space Dirac equation(s):

$$(\gamma^\mu p_\mu - mc)u = 0$$

their adjoints, $\bar{u} = u^\dagger \gamma^0$, $\bar{v} = v^\dagger \gamma^0$, satisfy

$$\bar{u}(\gamma^\mu p_\mu - mc) = 0$$

They are orthogonal,

$$\bar{u}^{(1)} u^{(2)} = 0$$

normalized,

$$\bar{u}u = 2mc$$

and *complete*, in the sense that

$$\sum_{s=1,2} u^{(s)} \bar{u}^{(s)} = (\gamma^\mu p_\mu + mc)$$

Positrons

$$\psi(x) = ae^{(i/\hbar)p \cdot x} v^{(s)}(p)$$

$$(\gamma^\mu p_\mu + mc)v = 0$$

$$\bar{v}(\gamma^\mu p_\mu + mc) = 0$$

$$\bar{v}^{(1)} v^{(2)} = 0$$

$$\bar{v}v = -2mc$$

$$\sum_{s=1,2} v^{(s)} \bar{v}^{(s)} = (\gamma^\mu p_\mu - mc)$$

Photons

$$A_\mu(x) = ae^{-(i/\hbar)p \cdot x} \epsilon_\mu^{(s)}$$

where $s = 1, 2$ for the two spin states (polarizations). The polarization vectors $\epsilon_\mu^{(s)}$ satisfy the momentum space Lorentz condition:

$$p^\mu \epsilon_\mu = 0$$

They are orthogonal, in the sense that

$$\epsilon_\mu^{(1)*} \epsilon^{(2)\mu} = 0$$

and normalized

$$\epsilon^{\mu*} \epsilon_\mu = -1$$

In the Coulomb gauge

$$\epsilon^0 = 0, \quad \epsilon \cdot \mathbf{p} = 0$$

and the polarization three-vectors obey the *completeness relation*

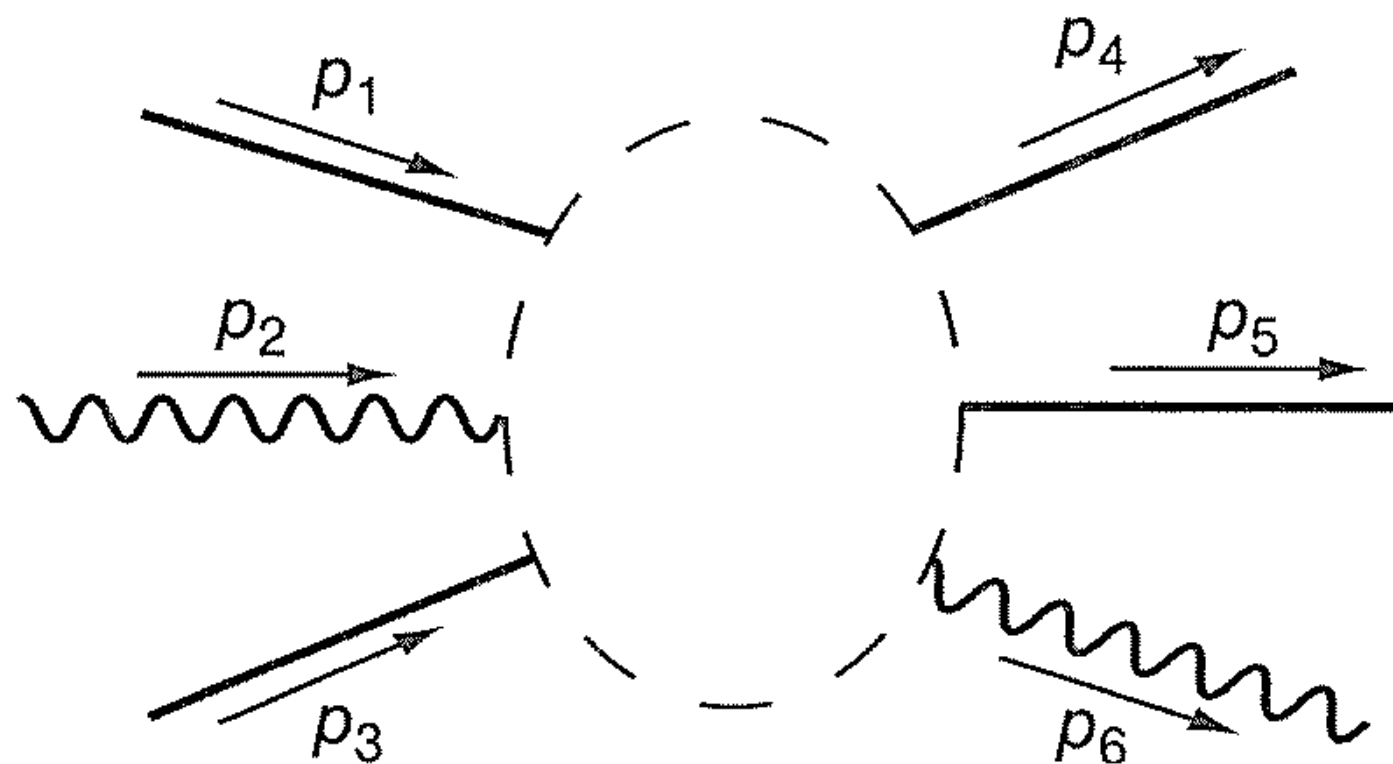
$$\sum_{s=1,2} \epsilon_i^{(s)} \epsilon_j^{(s)*} = \delta_{ij} - \hat{p}_i \hat{p}_j$$

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Feynman Rules

1. *Notation*: To each *external* line associate a momentum $p_1, p_2, \dots, p_n$, and draw an arrow next to the line, indicating the positive direction (forward in time).* To each *internal* line associate a momentum $q_1, q_2, \dots$; again draw an arrow next to the line indicating the positive direction (arbitrarily assigned).
2. *External lines*: External lines contribute factors as follows:

$$\begin{array}{lcl}
 \text{Electrons :} & \left\{ \begin{array}{l} \text{Incoming}(\rightarrow\bullet) : u \\ \text{Outgoing}(\bullet\rightarrow) : \bar{u} \end{array} \right. \\
 \text{Positrons :} & \left\{ \begin{array}{l} \text{Incoming}(\leftarrow\bullet) : \bar{v} \\ \text{Outgoing}(\bullet\leftarrow) : v \end{array} \right. \\
 \text{Photons :} & \left\{ \begin{array}{l} \text{Incoming}(\text{wavy}\bullet) : \epsilon_\mu \\ \text{Outgoing}(\bullet\text{wavy}) : \epsilon_\mu^* \end{array} \right.
 \end{array}$$



3. *Vertex factors*: Each vertex contributes a factor

$$ig_e \gamma^\mu$$

The dimensionless coupling constant g_e is related to the charge of the electron: $g_e = e\sqrt{4\pi/\hbar c} = \sqrt{4\pi\alpha}$.*

4. *Propagators*: Each internal line contributes a factor as follows:

$$\text{Electrons and positrons: } \frac{i(\gamma^\mu q_\mu + mc)}{q^2 - m^2 c^2}$$

$$\text{Photons: } \frac{-ig_{\mu\nu}}{q^2}$$

5. *Conservation of energy and momentum*: For each vertex, write a delta function of the form

$$(2\pi)^4 \delta^4(k_1 + k_2 + k_3)$$

where the k 's are the three four-momenta coming *into* the vertex (if an arrow leads *outward*, then k is *minus* the four-momentum of that line).

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Feynman Rules

6. *Integrate over internal momenta:* For each internal momentum q , write a factor

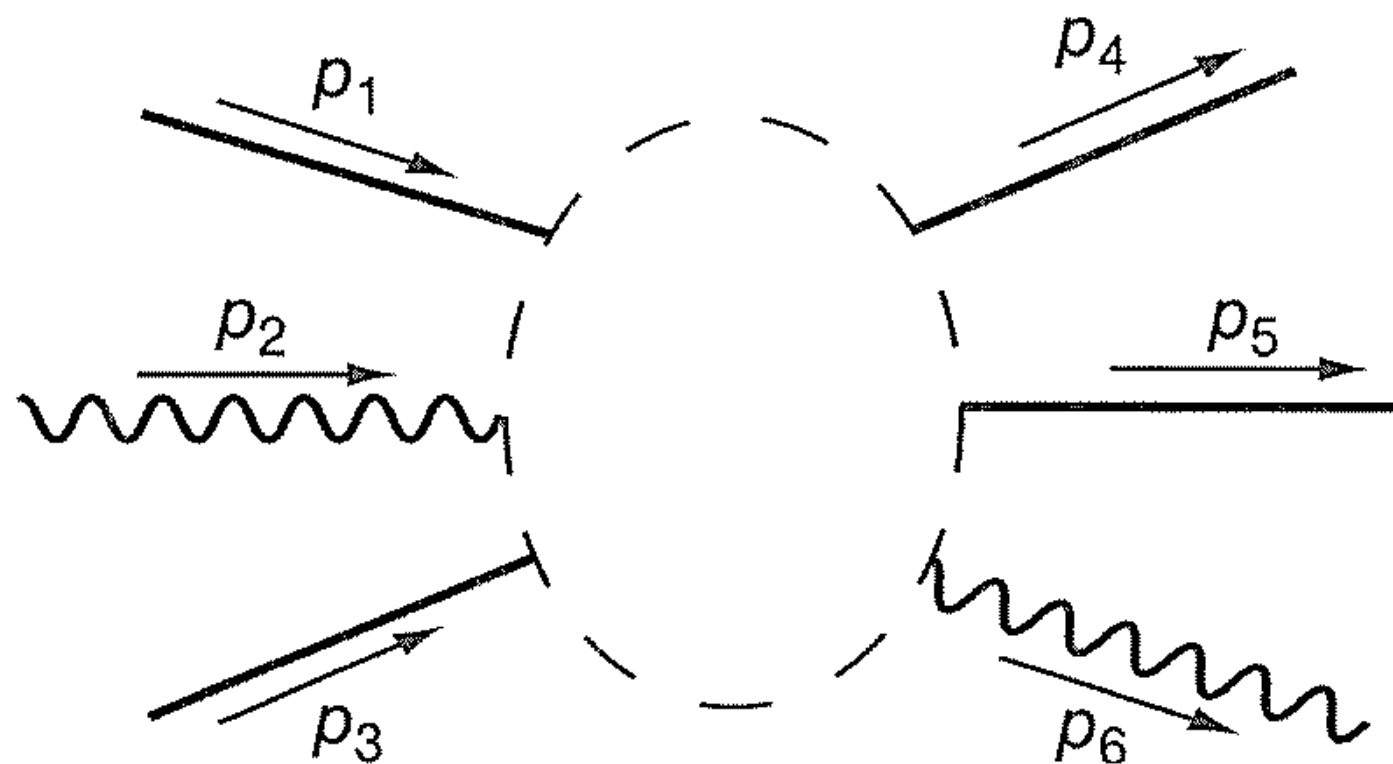
$$\frac{d^4 q}{(2\pi)^4}$$

and integrate.

7. *Cancel the delta function:* The result will include a factor

$$(2\pi)^4 \delta^4(p_1 + p_2 + \cdots - p_n)$$

corresponding to overall energy–momentum conservation. Cancel this factor, and multiply by i ; what remains is $\mathcal{M}$.



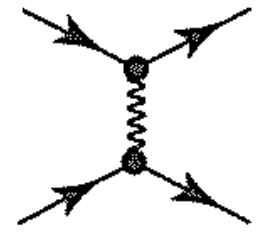
As before, the idea is to draw all the diagrams contributing to the process in question (up to the desired order), calculate the amplitude ($\mathcal{M}$) for each one, and add them up to get the *total* amplitude, which is then inserted into the appropriate Golden Rule for the decay rate or the scattering cross section, as the case may be. There's one new twist that occasionally arises: the antisymmetrization of fermion wave functions requires that we insert a minus sign in combining amplitudes that differ only in the interchange of two identical external fermions. It doesn't matter *which* diagram you associate the minus sign with, since the total will be squared eventually anyway; but there must be a *relative* minus sign between them:

8. *Antisymmetrization:* Include a minus sign between diagrams that differ only in the interchange of two incoming (or outgoing) electrons (or positrons), or of an incoming electron with an outgoing positron (or vice versa).

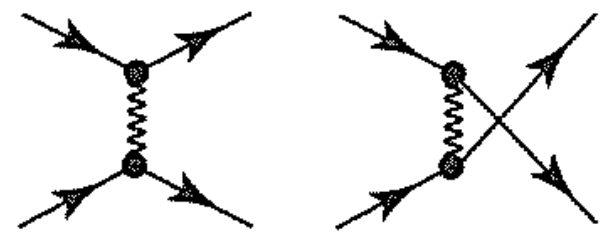
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Second-order processes

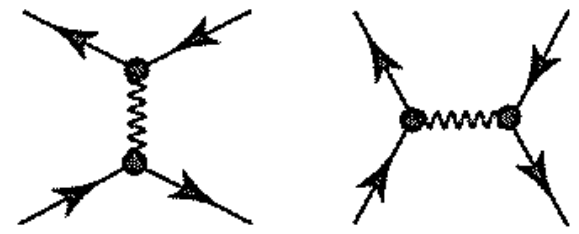
Elastic



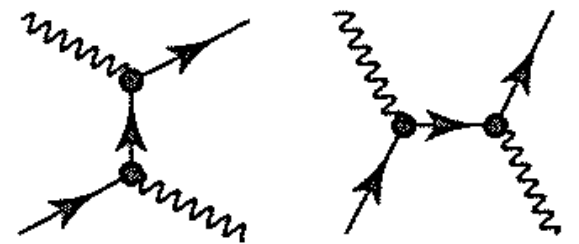
{ Electron–muon scattering($e + \mu \rightarrow e + \mu$)
(Mott scattering($M \gg m$) $\Rightarrow$ Rutherford scattering($v \ll c$))



{ Electron–electron scattering($e^- + e^- \rightarrow e^- + e^-$)
(Møller scattering)



{ Electron–positron scattering($e^- + e^+ \rightarrow e^- + e^+$)
(Bhabha scattering)



{ Compton scattering($\gamma + e^- \rightarrow \gamma + e^-$)

Inelastic

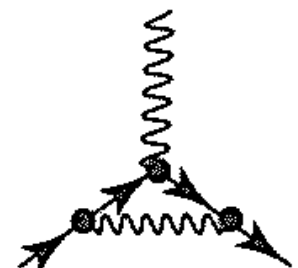


{ Pair annihilation($e^- + e^+ \rightarrow \gamma + \gamma$)



{ Pair production($\gamma + \gamma \rightarrow e^- + e^+$)

Most important third-order process



{ $\Rightarrow$ Anomalous magnetic moment of electron

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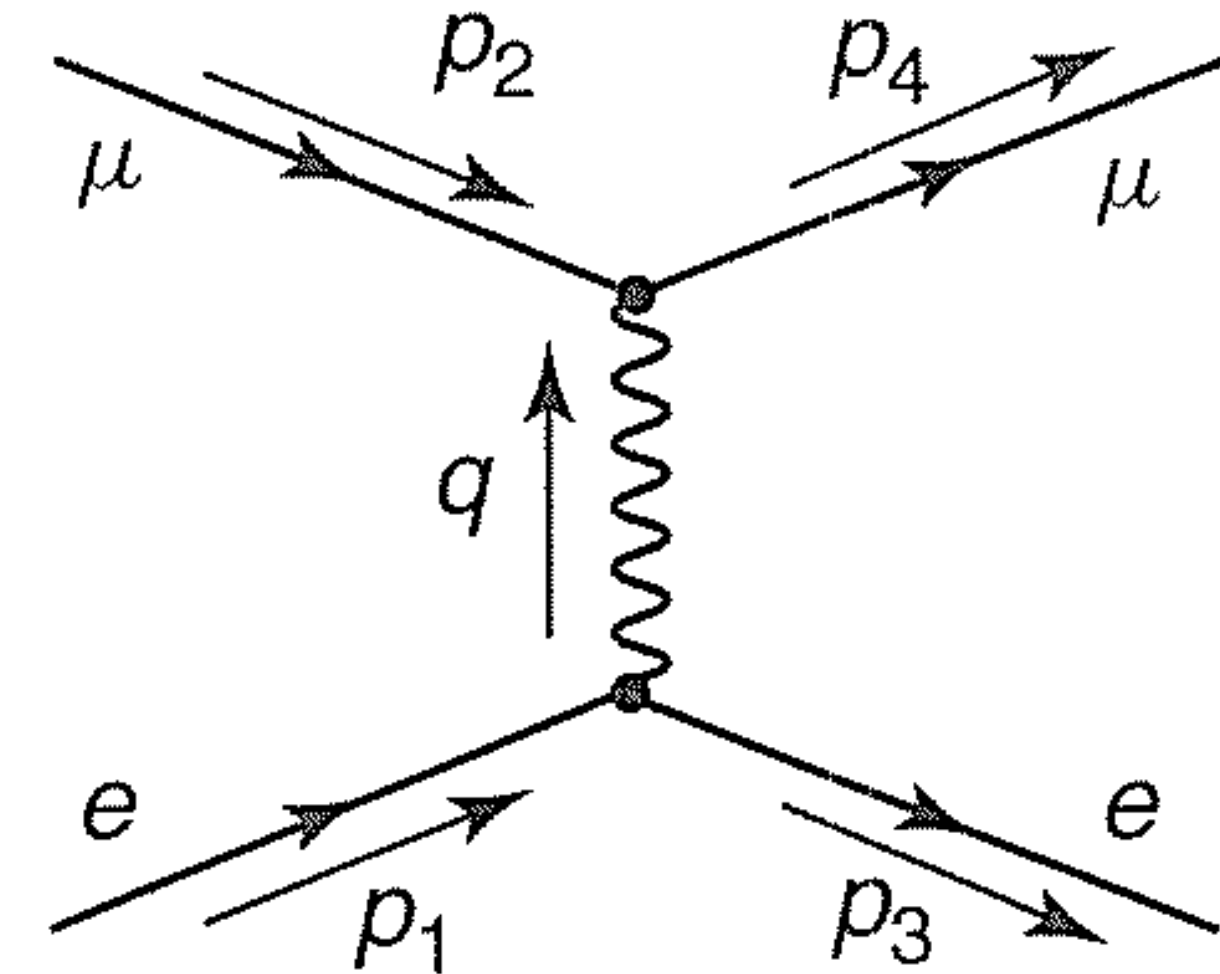
Example 7.1 Electron–Muon Scattering Walking ‘backward’ along each fermion line and applying the Feynman rules as we go:

$$(2\pi)^4 \int [\bar{u}^{(s_3)}(p_3)(ig_e\gamma^\mu)u^{(s_1)}(p_1)] \frac{-ig_{\mu\nu}}{q^2} [\bar{u}^{(s_4)}(p_4)(ig_e\gamma^\nu)u^{(s_2)}(p_2)] \\ \times \delta^4(p_1 - p_3 - q)\delta^4(p_2 + q - p_4) d^4q$$

Notice how the space-time indices on the photon propagator contract with those of the vertex factors at either end of the photon line. Carrying out the (trivial) q integration, and dropping the overall delta function, we find

$$\mathcal{M} = -\frac{g_e^2}{(p_1 - p_3)^2} [\bar{u}^{(s_3)}(p_3)\gamma^\mu u^{(s_1)}(p_1)][\bar{u}^{(s_4)}(p_4)\gamma_\mu u^{(s_2)}(p_2)]$$

In spite of its complicated appearance, with four spinors and eight γ matrices, this is just a *number*, which you can work out once the spins are specified



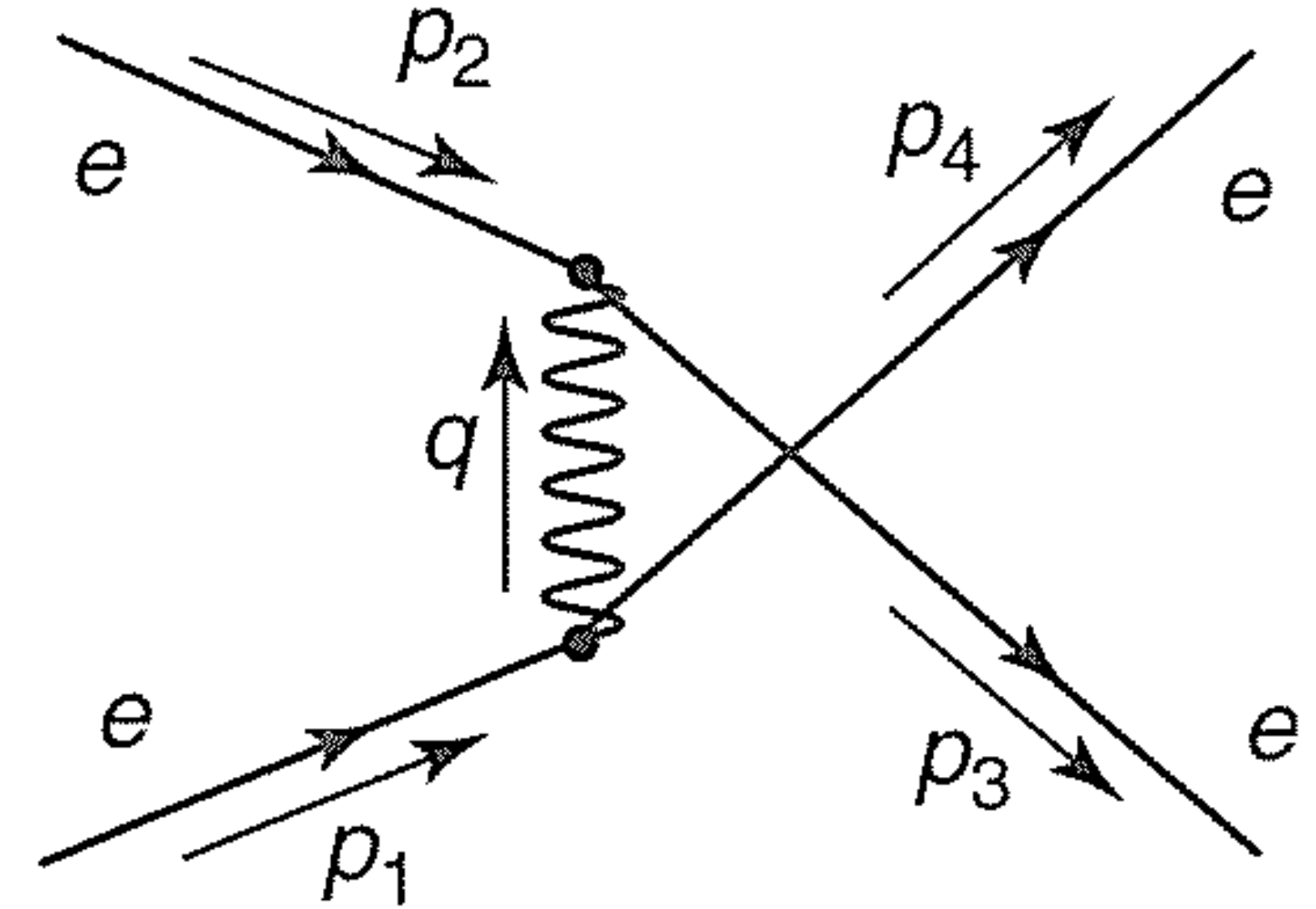
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Example 7.2 Electron–Electron Scattering

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Example 7.2 Electron–Electron Scattering In this case there is a second diagram, in which the electron that emerges with momentum p_3 and spin s_3 comes from the p_2, s_2 electron, instead of the p_1, s_1 electron. We can obtain this amplitude simply by the replacement $p_3, s_3 \leftrightarrow p_4, s_4$. According to Rule 8, the two diagrams are to be *subtracted*, so the total amplitude is

$$\begin{aligned}\mathcal{M} = & -\frac{g_e^2}{(p_1 - p_3)^2} [\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)] \\ & + \frac{g_e^2}{(p_1 - p_4)^2} [\bar{u}(4)\gamma^\mu u(1)][\bar{u}(3)\gamma_\mu u(2)]\end{aligned}$$



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Example 7.3 Electron–Positron Scattering Again, there are two diagrams.* The first is similar to the electron–muon diagram

$$(2\pi)^4 \int [\bar{u}(3)(ig_e\gamma^\mu)u(1)] \frac{-ig_{\mu\nu}}{q^2} [\bar{v}(2)(ig_e\gamma^\nu)v(4)] \\ \times \delta^4(p_1 - p_3 - q)\delta^4(p_2 + q - p_4) d^4q$$

Notice that ‘proceeding backwards’ along an *antiparticle* line means working *forward* in time; the order is always *adjoint/matrix/spinor*. The amplitude for this diagram is thus

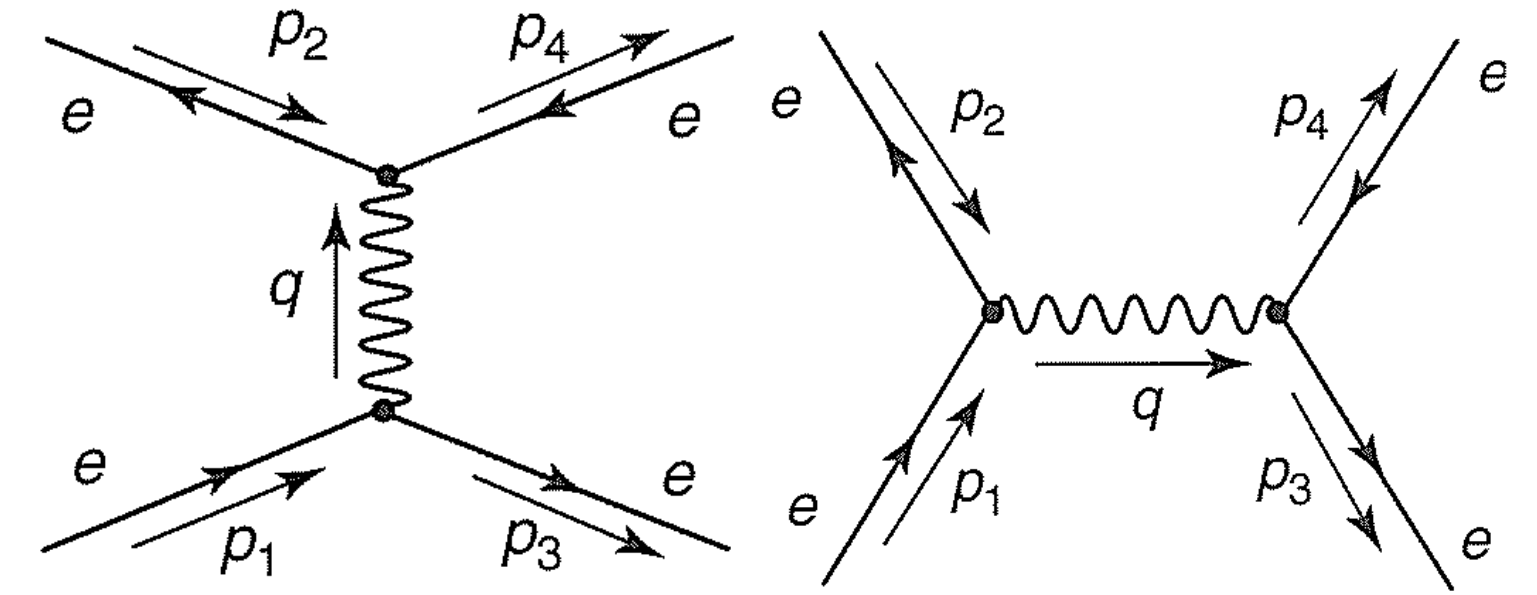
$$\mathcal{M}_1 = -\frac{g_e^2}{(p_1 - p_3)^2} [\bar{u}(3)\gamma^\mu u(1)][\bar{v}(2)\gamma_\mu v(4)]$$

The other diagram represents virtual annihilation of the electron and positron, followed by pair production

$$(2\pi)^4 \int [\bar{u}(3)(ig_e\gamma^\mu)v(4)] \frac{-ig_{\mu\nu}}{q^2} [\bar{v}(2)(ig_e\gamma^\nu)u(1)] \\ \times \delta^4(q - p_3 - p_4)\delta^4(p_1 + p_2 - q) d^4q$$

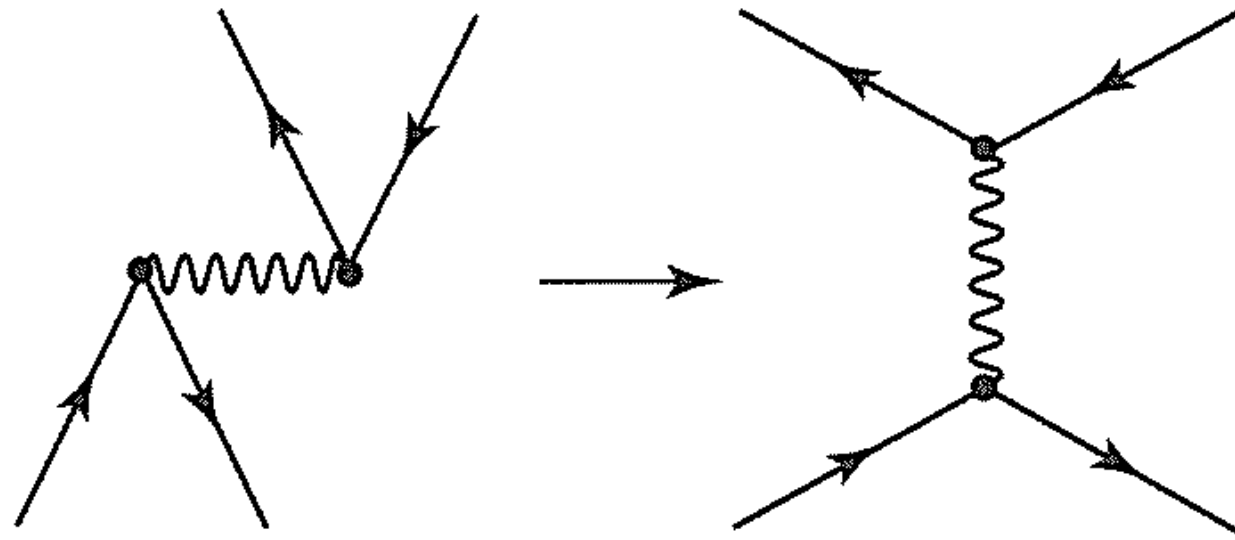
The amplitude for this diagram is therefore

$$\mathcal{M}_2 = -\frac{g_e^2}{(p_1 + p_2)^2} [\bar{u}(3)\gamma^\mu v(4)][\bar{v}(2)\gamma_\mu u(1)]$$



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Now, should we *add* these amplitudes, or *subtract* them? Interchanging the incoming positron and the outgoing electron in the second diagram and then redrawing it in a more customary configuration

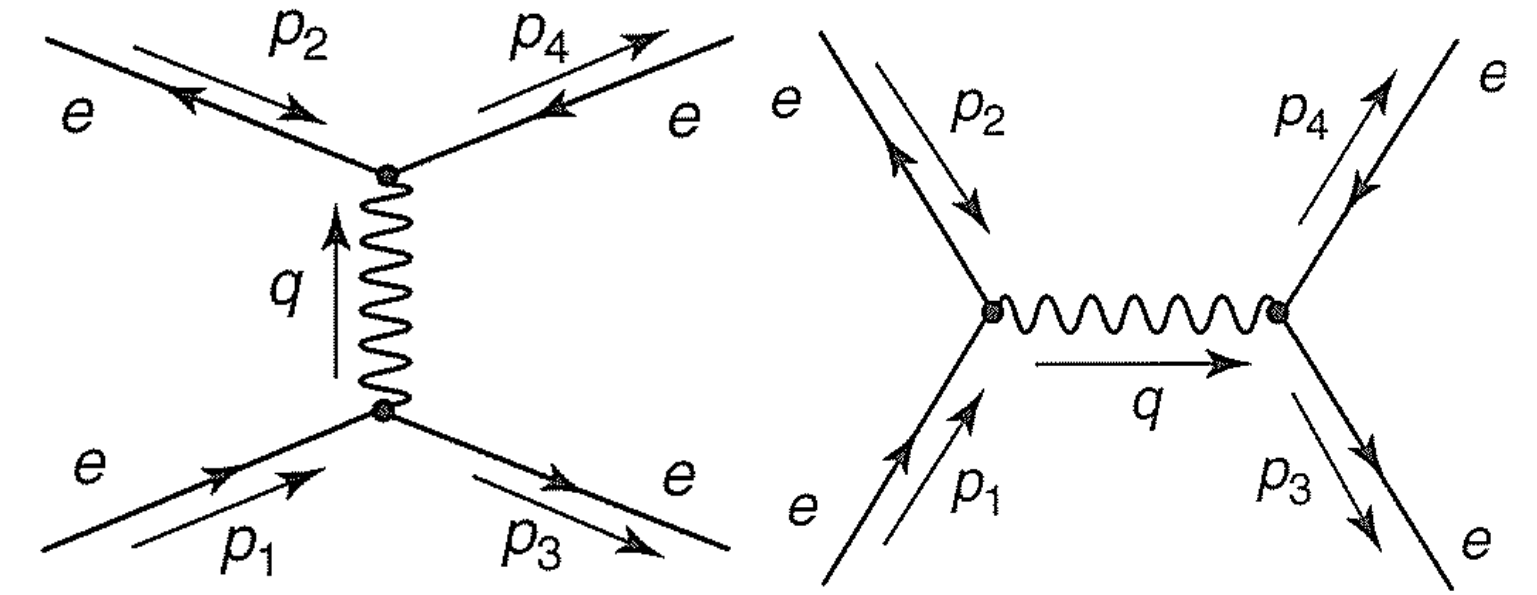


we recover the first diagram minus sign:

According to Rule 8, then, we need a

$$\mathcal{M} = -\frac{g_e^2}{(p_1 - p_3)^2} [\bar{u}(3)\gamma^\mu u(1)][\bar{v}(2)\gamma_\mu v(4)]$$

$$+ \frac{g_e^2}{(p_1 + p_2)^2} [\bar{u}(3)\gamma^\mu v(4)][\bar{v}(2)\gamma_\mu u(1)]$$



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Casimir's Trick

In some experiments the incoming and outgoing electron (or positron) spins are specified, and the photon polarizations are given. If so, the next thing to do is to insert the appropriate spinors and polarization vectors into the expression for $\mathcal{M}$, and compute $|\mathcal{M}|^2$ – the quantity we actually need, to determine cross sections and lifetimes. More often, however, we are not interested in the spins. A typical experiment starts out with a beam of particles whose orientations are random, and simply counts the *number* of particles scattered in a given direction. In this case the relevant cross section is the *average* over all initial spin configurations, s_i , and the *sum* over all final spin configurations, s_f . In principle, we could compute $|\mathcal{M}(s_i \rightarrow s_f)|^2$ for every possible combination, and *then* do the summing and averaging:

$$\langle |\mathcal{M}|^2 \rangle \equiv \text{average over initial spins, sum over final spins,} \\ \text{of } |\mathcal{M}(s_i \rightarrow s_f)|^2 \quad (7.114)$$

In practice, it is much easier to compute $\langle |\mathcal{M}|^2 \rangle$ directly, without ever evaluating the individual amplitudes.

Consider, for instance, the electron–muon scattering amplitude (Equation 7.106). Squaring, we have

$$|\mathcal{M}|^2 = \frac{g_e^4}{(p_1 - p_3)^4} [\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)][\bar{u}(3)\gamma^\nu u(1)]^* [\bar{u}(4)\gamma_\nu u(2)]^* \quad (7.115)$$

(I use ν for the second contraction, since μ has been preempted.) A glance at the first and third ‘sandwiches’ (or the second and fourth) reveals that we must handle

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Casimir's Trick

quantities of the generic form

$$G \equiv [\bar{u}(a)\Gamma_1 u(b)][\bar{u}(a)\Gamma_2 u(b)]^*$$

where a and b stand for the appropriate spins and momenta, and Γ_1 , and Γ_2 are 4×4 matrices. All the other processes described – Møller, Bhabha, and Compton scattering, as well as pair production and annihilation – lead to expressions with similar structure. To begin with, we evaluate the complex conjugate (which is the same as the Hermitian conjugate, since the quantity in square brackets is a 1×1 ‘matrix’):

$$[\bar{u}(a)\Gamma_2 u(b)]^* = [u(a)^\dagger \gamma^0 \Gamma_2 u(b)]^\dagger = u(b)^\dagger \Gamma_2^\dagger \gamma^{0\dagger} u(a)$$

Now, $\gamma^{0\dagger} = \gamma^0$, and $(\gamma^0)^2 = 1$, so

$$u(b)^\dagger \Gamma_2^\dagger \gamma^{0\dagger} u(a) = u(b)^\dagger \gamma^0 \gamma^0 \Gamma_2^\dagger \gamma^0 u(a) = \bar{u}(b) \bar{\Gamma}_2 u(a)$$

where*

$$\bar{\Gamma}_2 \equiv \gamma^0 \Gamma_2^\dagger \gamma^0$$

Thus

$$G = [\bar{u}(a)\Gamma_1 u(b)][\bar{u}(b)\bar{\Gamma}_2 u(a)]$$

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Casimir's Trick

We are now ready to sum over the spin orientations of particle b . Using the completeness relation, we have

$$\begin{aligned}\sum_{b \text{ spins}} G &= \bar{u}(a)\Gamma_1 \left\{ \sum_{s_b=1,2} u^{(s_b)}(p_b)\bar{u}^{(s_b)}(p_b) \right\} \bar{\Gamma}_2 u(a) \\ &= \bar{u}(a)\Gamma_1(\not{p}_b + m_b c)\bar{\Gamma}_2 u(a) = \bar{u}(a)Qu(a)\end{aligned}$$

where Q is a temporary shorthand for the 4×4 matrix

$$Q \equiv \Gamma_1(\not{p}_b + m_b c)\bar{\Gamma}_2$$

Next, we do the same for particle a :

$$\sum_{a \text{ spins}} \sum_{b \text{ spins}} G = \sum_{s_a=1,2} \bar{u}^{(s_a)}(p_a)Qu^{(s_a)}(p_a)$$

Or, writing out the matrix multiplication explicitly:*

$$\begin{aligned}\sum_{s_a=1,2} \sum_{i,j=1}^4 \bar{u}^{(s_a)}(p_a)_i Q_{ij} u^{(s_a)}(p_a)_j &= \sum_{i,j=1}^4 Q_{ij} \left\{ \sum_{s_a=1,2} u^{(s_a)}(p_a)\bar{u}^{(s_a)}(p_a) \right\}_{ji} \\ &= \sum_{i,j=1}^4 Q_{ij}(\not{p}_a + m_a c)_{ji} = \sum_{i=1}^4 [Q(\not{p}_a + m_a c)]_{ii} = \text{Tr}[Q(\not{p}_a + m_a c)]\end{aligned}$$

where ‘Tr’ denotes the *trace* of the matrix (the sum of its diagonal elements)

Conclusion:

$$\sum_{\text{all spins}} [\bar{u}(a)\Gamma_1 u(b)][\bar{u}(a)\Gamma_2 u(b)]^* = \text{Tr}[\Gamma_1(\not{p}_b + m_b c)\bar{\Gamma}_2(\not{p}_a + m_a c)]$$

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Casimir's Trick

Example 7.5 In the case of electron–muon scattering $\Gamma_2 = \gamma^\nu$,
and hence $\bar{\Gamma}_2 = \gamma^0 \gamma^{\nu\dagger} \gamma^0 = \gamma^\nu$ Applying Casimir's trick twice, we
find

$$\begin{aligned} \langle |\mathcal{M}|^2 \rangle &= \frac{g_e^4}{4(p_1 - p_3)^4} \text{Tr}[\gamma^\mu (\not{p}_1 + mc) \gamma^\nu (\not{p}_3 + mc)] \\ &\quad \times \text{Tr}[\gamma_\mu (\not{p}_2 + Mc) \gamma_\nu (\not{p}_4 + Mc)] \end{aligned}$$

where m is the mass of the electron and M is the mass of the muon. The factor of $\frac{1}{4}$ is included because we want the *average* over the initial spins; since there are two particles, each with two possible orientations, the average is a quarter of the sum.

Casimir's trick reduces everything to an exercise in calculating the trace of some complicated product of γ matrices. This algebra is facilitated by a number of theorems, which I shall now list

First of all, I should mention three facts about traces in general: if A and B are any two matrices, and α is any number

1. $\text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$
2. $\text{Tr}(\alpha A) = \alpha \text{Tr}(A)$
3. $\text{Tr}(AB) = \text{Tr}(BA)$

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It follows from number 3 that $\text{Tr}(ABC) = \text{Tr}(CAB) = \text{Tr}(BCA)$, but this is *not* equal, in general, to the trace of the matrices taken in the other order: $\text{Tr}(ACB) = \text{Tr}(BAC) = \text{Tr}(CBA)$. You can ‘peel’ matrices off the back end of a product and move them around to the front, but you must preserve the ordering. It is useful to note that

$$4. \quad g_{\mu\nu} g^{\mu\nu} = 4$$

and to recall the fundamental anticommutation relation for the γ matrices (together with an associated rule for ‘slash’ products):

$$5. \quad \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu} \quad 5'. \quad \not{a} \not{b} + \not{b} \not{a} = 2a \cdot b$$

From these there follows a sequence of ‘contraction theorems’:

$$6. \quad \gamma_\mu \gamma^\mu = 4$$

$$7. \quad \gamma_\mu \gamma^\nu \gamma^\mu = -2\gamma^\nu \quad 7'. \quad \gamma_\mu \not{a} \gamma^\mu = -2\not{a}$$

$$8. \quad \gamma_\mu \gamma^\nu \gamma^\lambda \gamma^\mu = 4g^{\nu\lambda} \quad 8'. \quad \gamma_\mu \not{a} \not{b} \gamma^\mu = 4(a \cdot b)$$

$$9. \quad \gamma_\mu \gamma^\nu \gamma^\lambda \gamma^\sigma \gamma^\mu = -2\gamma^\sigma \gamma^\lambda \gamma^\nu \quad 9'. \quad \gamma_\mu \not{a} \not{b} \not{c} \gamma^\mu = -2\not{c} \not{b} \not{a}$$

and a collection of ‘trace theorems’:

$$10. \quad \text{The trace of the product of an odd number of gamma matrices is zero.}$$

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$$11. \quad \text{Tr}(1) = 4$$

$$12. \quad \text{Tr}(\gamma^\mu \gamma^\nu) = 4g^{\mu\nu} \quad 12'. \quad \text{Tr}(\not{a} \not{b}) = 4(a \cdot b)$$

$$13. \quad \text{Tr}(\gamma^\mu \gamma^\nu \gamma^\lambda \gamma^\sigma) = 4(g^{\mu\nu} g^{\lambda\sigma} - g^{\mu\lambda} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\lambda}) \quad 13'. \quad \text{Tr}(\not{a} \not{b} \not{c} \not{d}) = 4(a \cdot b c \cdot d - a \cdot c b \cdot d + a \cdot d b \cdot c)$$

Finally, since $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ is the product of an *even* number of γ matrices, it follows from Rule 10 that $\text{Tr}(\gamma^5 \gamma^\mu) = \text{Tr}(\gamma^5 \gamma^\mu \gamma^\nu \gamma^\lambda) = 0$. When γ^5 is multiplied by an *even* number of γ 's, we find

$$14. \quad \text{Tr}(\gamma^5) = 0$$

$$15. \quad \text{Tr}(\gamma^5 \gamma^\mu \gamma^\nu) = 0 \quad 15'. \quad \text{Tr}(\gamma^5 \not{a} \not{b}) = 0$$

$$16. \quad \text{Tr}(\gamma^5 \gamma^\mu \gamma^\nu \gamma^\lambda \gamma^\sigma) = 4i\epsilon^{\mu\nu\lambda\sigma} \quad 16'. \quad \text{Tr}(\gamma^5 \not{a} \not{b} \not{c} \not{d}) = 4i\epsilon^{\mu\nu\lambda\sigma} a_\mu b_\nu c_\lambda d_\sigma$$

where*

$$\epsilon^{\mu\nu\lambda\sigma} \equiv \begin{cases} -1, & \text{if } \mu\nu\lambda\sigma \text{ is an } \textit{even} \text{ permutation of } 0123, \\ +1, & \text{if } \mu\nu\lambda\sigma \text{ is an } \textit{odd} \text{ permutation,} \\ 0, & \text{if any two indices are the same.} \end{cases}$$